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Efficacy and adverse events of oral isotretinoin for acne: a systematic review*

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Summary

Despite many years of clinical use of isotretinoin, a comprehensive review of evidence for isotretinoin therapy in patients with acne is lacking. We searched MEDLINE, Embase, Cochrane Central, relevant web pages and bibliographies for randomized controlled trials in acne evaluating isotretinoin vs. control (placebo or other therapy). Data were extracted and summarized descriptively. Eleven trials were identified (total 760 patients randomized), containing mostly men. Mean treatment ages ranged from 18 to 47·9 years and participants generally had moderate-to-severe acne. Across all trials, isotretinoin therapy reduced acne lesion counts by a clinically relevant amount, and always by a greater amount than control, which was either placebo (two studies), oral antibiotics (seven studies) or other control (two studies). Across trials with an overall low risk of bias, two of three demonstrated statistically significant differences between isotretinoin and control. The frequency of adverse events was twice as high with isotretinoin (751 events) than with control (388 events). More than half of all adverse events were dermatological and related to dryness. Adverse events from isotretinoin causing participant withdrawal from trials (12 patients) included Stevens–Johnson syndrome, cheilitis, xerosis, acne flare, photophobia, elevated liver enzymes, decreased appetite, headaches and depressed mood. This review suggests that isotretinoin is effective in reducing acne lesion counts, but adverse events are common. This study was registered with PROSPERO number CRD42015025080.

What's already known about this topic?

- Isotretinoin is used for management of moderate-to-severe acne.
- It is considered effective and generally safe, although a comprehensive review of evidence from randomized controlled trials has not been performed.

What does this study add?

- This study reviewed evidence from randomized controlled trials to assess the efficacy and safety profile of isotretinoin for treatment of acne.
- It was found that isotretinoin was superior to placebo and other therapies in reducing acne lesion counts; however, isotretinoin also had far more adverse events.
- While these adverse events were generally mild and dryness related, severe adverse events requiring participant withdrawal occurred in 3·2% of patients randomized to isotretinoin.

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Acne vulgaris is the most common skin disease, and the associated anxiety, reduced self-esteem, stigma and facial scarring can impose a significant burden of disease on the patient.[1–5] In cases of severe nodulocystic and papulopustular acne, oral isotretinoin is often prescribed as a first-line treatment,[6,7] potentially resulting in cure of disease.[8,9] Recent Canadian, American and European clinical practice guidelines have suggested that isotretinoin should be used for management of moderate-to-severe acne.[10–12]

Various narrative reviews on acne have discussed the efficacy and adverse events of oral isotretinoin, but a comprehensive systematic review of outcomes and study quality has surprisingly not been performed.[6,13,14] One meta-analysis was conducted in 1999 on isotretinoin effectiveness;[15] however, this study considered single-arm studies without comparators, did not evaluate treatment based on lesion counts, did not consider study quality or risk of bias and did not report adverse events. Systematic reviews have been published on isotretinoin adverse events such as psychiatric outcomes,[16] laboratory abnormalities[17] and inflammatory bowel disease;[18] however, these were either not focused on randomized controlled trial evidence and thus are prone to confounding by indication, or did not concurrently consider the effect of isotretinoin. Therefore, given the prevalence of acne and broad use of isotretinoin, a review focusing on the highest quality of evidence on treatment efficacy and adverse events is well justified.

The objective of this systematic review was to summarize results from all available randomized clinical trials on oral isotretinoin to compare its clinical efficacy and adverse event profile with placebo or alternative therapies.

Methods

This systematic review was conducted in accordance with the PRISMA guidelines,[19] and has been registered with PROSPERO (CRD42015025080). We searched the MEDLINE, Embase and Cochrane Central databases using key words related to acne and isotretinoin without language restrictions, and manually searched study reference lists and clinical trial registries up to 18 October 2016. Following duplicate removal, two authors (I.A.V. and R.T.L) screened titles and abstracts to identify studies that reported randomized controlled trials of isotretinoin for acne. The kappa statistic was used to determine percentage agreement (with 95% confidence interval) between reviewers. The primary outcome of interest was treatment efficacy of oral isotretinoin, classified as change in acne lesion counts where possible. χ^2-Tests were used to compare percentage changes in primary outcome measures between isotretinoin and control groups ($\alpha = 0{\cdot}05$). To establish the clinical relevance of statistical tests, the recommendation from the Canadian clinical practice guideline on acne was used, where clinical relevance is defined as a reduction in acne lesion counts by > 10%.[10,20]

Secondary outcomes in this review included adverse events from acne therapy. Study quality was assessed by the same two authors using the Cochrane Collaboration's Risk of Bias Assessment Tool,[21] and any disagreements were resolved by consensus-based discussion. We summarized data through descriptive tables and narrative discussion, placing emphasis on the highest-quality trials, which we defined as those with a low risk of bias across criteria that would have the strongest impact on efficacy or adverse event data. Full details of our search strategy, methodology for data extraction, risk-of-bias assessment and outcomes extraction are described in Appendix S1 (see Supporting Information).

Results

In total 1237 studies were identified, of which 460 were duplicates. Forty articles were identified for full-text review, of which 11 satisfied the inclusion criteria and were considered for qualitative synthesis (Fig. 1).[22–33] Through this process, we achieved a kappa statistic of 89·9% (95% confidence interval 75·7–100), demonstrating good agreement among authors regarding which studies to include.

Only one study was classified as having a low risk of bias across all nine criteria considered in study quality (Fig. 2).[29] By focusing on only the criteria with the greatest influence on efficacy and adverse event outcomes, three studies were

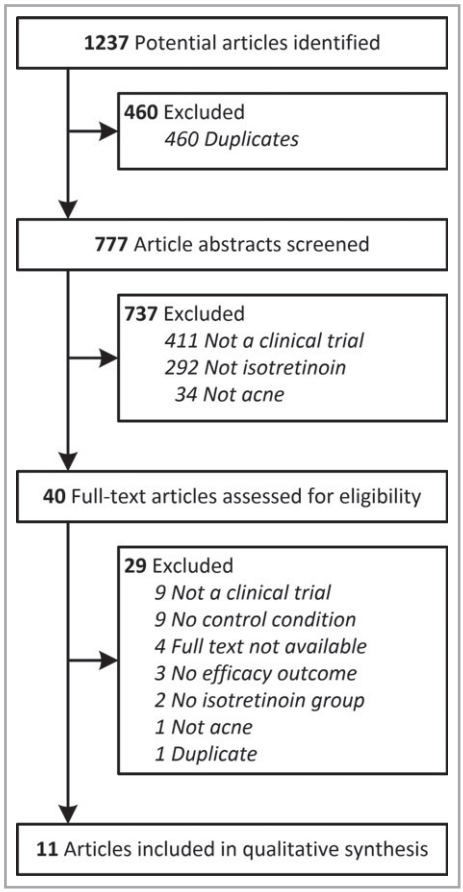

Fig 1. Flow diagram for studies included in the review. We searched MEDLINE, Embase, Cochrane Central, trial web pages and article reference lists for eligible studies. These were screened to identify 11 trials meeting eligibility for review.

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classified as having an overall low risk of bias, three had an unclear risk of bias and five had a high risk of bias. A summary of study characteristics for each trial included in this review is provided in Table 1. In total, 760 participants were randomized, and outcome data were available for 748 patients. Of the 760 randomized, only 156 were female. Across trials, the mean treatment age ranged from 18·0 to 47·9 years, and most studies classified participants as having moderate-to-severe acne, usually based on lesion counts (and combinations of papules, pustules and nodules) or grading scales. Nine of the 11 studies evaluated moderate-to-severe nodular or cystic acne (Table 1), which is consistent with the U.S. Food and Drug Administration approval for isotretinoin; however, there was no consistent approach to quantify this definition of acne severity.

Treatment efficacy

Where possible, we report treatment efficacy data based on the last available time point at which both treatment and control groups received the assigned intervention. This is the time point provided in Table 1 (and does not necessarily match the study's total duration). This last time point was chosen for reporting treatment efficacy to ensure that comparisons were made only during periods of active treatment. Based on this point of comparison, all trials reported clinically relevant (> 10%)[10,20] reductions in lesion counts with use of isotretinoin (Table 2). In addition, superior treatment efficacy of isotretinoin vs. control (> 10% difference, Table 2) was found in all trials, with seven of 11 trials showing statistical significance.

Placebo control

Only two trials were placebo controlled, with a total of 91 participants randomized (46 in the placebo control groups).[29,31] One trial had a short study period of 4 weeks,[29] while the other evaluated therapy for 16 weeks.[31] Participant baseline characteristics such as age, sex and acne severity were not clearly reported in one study,[29] whereas the other reported treatment on mostly female patients in their late 30s.[31] Both studies required pretreatment washout of previous therapies for acne. The placebo groups generally experienced no improvement or worsening in total lesion count, while the isotretinoin groups experienced reductions in total lesion counts of 32·0% and 69·8%. In both studies, differences in lesion count reductions were statistically significant. One of the placebo-controlled studies was classified as having an overall low risk of bias, and was the original trial comparing isotretinoin with placebo.[29]

Oral antibiotic control

Seven trials were performed with oral antibiotic comparators, comprising 593 participants randomized (304 in the antibiotic control groups). Among these trials, the antibiotics tested against isotretinoin included minocycline (n = 1),[23] erythromycin (n = 1),[24,25] tetracycline (n = 2),[26,28] dapsone (n = 1),[30] doxycycline (n = 1)[32] and azithromycin (n = 1).[33] Some studies also gave topical medications to patients in the antibiotic groups (Table 1). In each of these trials, isotretinoin reduced acne severity more than oral antibiotic therapy (Table 2); however, this was statistically significant for only

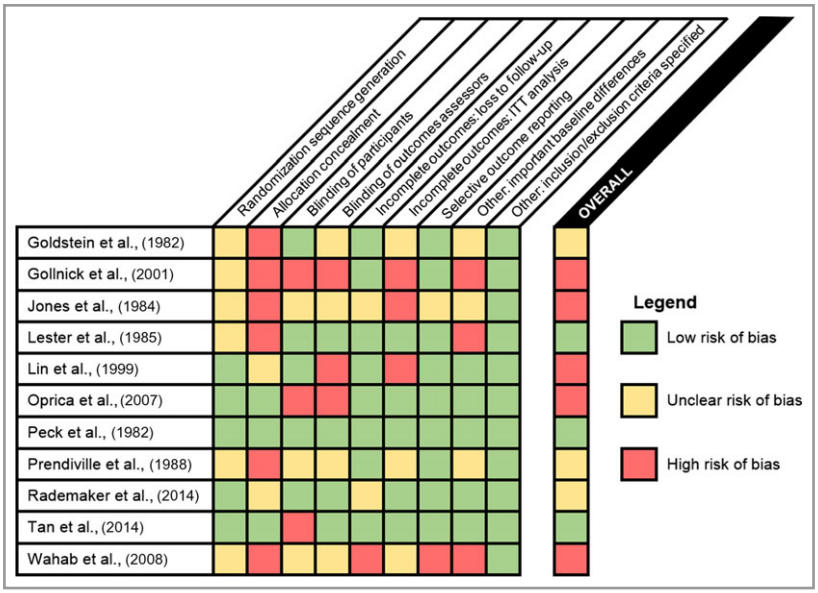

Fig 2. Study quality assessment. Study quality was assessed using the Cochrane Collaboration's Risk of Bias Assessment Tool according to specific study features. Each study was assigned either high, low or unclear risk of bias. An overall assessment of bias was performed based on the criteria that had the strongest impact on efficacy and adverse events data: (i) blinding of outcomes assessors; (ii) incomplete outcomes: loss to follow-up; (iii) incomplete outcomes: intention-to-treat (ITT) analysis; and (iv) selective outcome reporting.

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Table 1 Study characteristics of the included intervention studies for acne examining the effects of isotretinoin vs. a control group. Treatment duration is listed as duration of active treatment where isotretinoin was directly compared with a control intervention

Study	Country	Total randomized (female)	Mean age (years)	Treatment duration	Baseline acne severity	Oral isotretinoin (ISO) group	Control (CON) group
Goldstein 1982[22]	U.S.A.	56 (0)	23·4 (range 14–54)	8 weeks	Severe, treatment-resistant nodulocystic	1·0 mg kg^{-1} daily	Etretinate 1·0 mg kg^{-1} PO daily
Gollnick 2001[23]	Germany	85 (0)	ISO: 19·0 (range 15–27). CON: 19·0 (range 16–31)	6 months	Nodular papulopustular or conglobata. ISO: median lesion count = 72. CON: median lesion count = 88	0·8 mg kg^{-1} daily month 1, 0·7 mg kg^{-1} daily month 2, 0·5–0·7 mg kg^{-1} daily month 3, 0·5 mg kg^{-1} daily months 4–6 to cumulative dose of 106–112 mg kg^{-1}	Minocycline 50 mg PO BID plus 20% azelaic acid cream BID
Jones 1984[24,25]	U.K.	60 (19)	ISO: 24·0 (range 14–50). CON: 22 (range 14–43)	32 weeks	Severe nodulocystic. ISO: mean lesion count = 64. Control: mean lesion count = 76	0·5 mg kg^{-1} per day	Erythromycin 1000 mg PO daily
Lester 1985[26]	Canada	30 (1)	ISO: 24·3 (range 17–35). CON: 26·5 (range 18–37)	16 weeks	Severe nodulocystic. ISO: mean lesion count = 48. CON: mean lesion count = 40. Both with at least 10 deep lesions	1·0 mg kg^{-1} per day. Increments of 0·5 mg kg^{-1} allowed at biweekly intervals to maximum of 2·0 mg kg^{-1} per day	Tetracycline 500 mg kg^{-1} PO daily. Increments of 0·5 mg kg^{-1} allowed at biweekly intervals to maximum 2·0 mg kg^{-1} per day
Lin 1999[27]	Taiwan	20 (2)	ISO: 45·4 (range 24–63). CON: 47·9 (range 26–63)	3 months	Severe nodulocystic. Acne severity scale 4·0 for both groups	5·0 mg BID	Vitamin B complex 10·0 mg PO BID
Oprica 2007[28]	Sweden	52 (17)	ISO: median 18. CON: median 19	24 weeks	Moderate-to-severe papulopustular with nodules or conglobate. ISO: mean lesion count = 178. CON: mean lesion count = 177	0·5 mg kg^{-1} BID	Tetracycline hydrochloride 500 mg PO daily before meals plus topical 0·1% adapalene gel daily
Peck 1982[29]	U.S.A.	33 (unknown)	Unknown	4 weeks	Treatment-resistant cystic/conglobata. At least 10 inflamed deep dermal or subcutaneous acne cysts or nodules for each participant	0·5 mg kg^{-1} daily, increasing as needed	Placebo
Prendiville 1988[30]	U.K.	40 (0)	Unknown	16 weeks	Nodulocystic. ISO: mean lesion count = 63. CON: mean lesion count = 60	40 mg daily	Dapsone 100 mg PO daily

(continued)

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Table 1 (continued)

Study	Country	Total randomized (female)	Mean age (years)	Treatment duration	Baseline acne severity	Oral isotretinoin (ISO) group	Control (CON) group
Rademaker 2014[31]	New Zealand	58 (51)	ISO: 37·6 ± 8·0. CON: 38·5 ± 7·1	16 weeks	Low grade. ISO: mean lesion count = 11. CON: mean lesion count = 10	5 mg daily	Placebo
Tan 2014[32]	Canada	266 (39)	ISO: 19·3 ± 4·5. CON: 19·5 ± 5·0	20 weeks	Severe nodular. ISO: mean lesion count = 110. CON: mean lesion count = 104	0·5 mg kg^{-1} daily weeks 0–4, 1·0 mg kg^{-1} daily weeks 5–20	Doxycycline 200 mg PO plus adapalene 0·1%/benzoyl peroxide 2·5% gel daily
Wahab 2008[33]	Pakistan	60 (27)	ISO: 21·0 ± 4·2. CON: 21·5 ± 4·2	3 months	Moderate-to-severe. ISO: 10 with severe acne. CON: 5 with severe acne	0·5–1·0 mg kg^{-1} daily	Azithromycin 500 mg PO three times weekly

PO, orally; BID, twice daily.

three of seven antibiotics trials. These studies generally had similar participant ages (means of 18–24 years), extended treatment durations (16–24 weeks) and patients with severe acne at baseline based on lesion counts or grading scales. Among the two antibiotic studies with an overall low risk of bias, one demonstrated clinical relevance and statistical significance of isotretinoin vs. doxycycline plus topical 0·1% adapalene/2·5% benzoyl peroxide gel,[32] while the other demonstrated clinical relevance but not statistical significance of isotretinoin vs. tetracycline.[26]

Other control

In the remaining two trials, one was performed with an alternative retinoid comparator (etretinate),[22] and the other with a vitamin B complex comparator, for a total of 76 participants randomized (38 in the control groups).[27] While Lin *et al.* described vitamin B complex as a placebo,[27] recent evidence suggests that vitamin B may have an influence on acnegenesis, and so this study was categorized in this review as 'other control'.[34] In both the vitamin B and etretinate studies, isotretinoin reduced acne severity more than control (Table 2), and this difference was statistically significant. Importantly, the vitamin B trial was conducted on an older population with chronic kidney disease who were being treated concurrently with dialysis.

Adverse events

While many studies planned to monitor laboratory abnormalities, only six of the 11 trials planned a priori to monitor patients for any adverse event.[22,23,27,28,31,32] Table 3 shows adverse event frequencies, categorized by system affected. Here it can be seen that the total adverse event frequency was about twice as high in the isotretinoin groups than in the controls. Overall, approximately two adverse events occurred per patient receiving isotretinoin and approximately one adverse event occurred per control patient. In total, 12 of 372 (3·2%) participants randomized to isotretinoin withdrew from trials due to adverse events, while seven of 388 (1·8%) participants randomized to control groups withdrew due to adverse events. While the studies were too heterogeneous to perform meta-analysis, the three studies reporting the highest rates of adverse events (cheilitis most commonly) also used the highest doses of isotretinoin (1·0 mg kg^{-1} per day).[22,26,32]

Abnormal blood work

Reports of abnormal blood work were more frequent among isotretinoin groups than in controls, but the overall frequency of a blood work abnormality was low for both isotretinoin and control. Overall, abnormal blood work represented 15 of 751 (2·0%) adverse events for those treated with isotretinoin, most commonly elevated serum lipids or elevated liver enzymes. Notably, there were no reported cases of pancreatitis secondary to hypertriglyceridaemia. Two patients (0·5%)

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Table 2 Summary of treatment efficacy data from each intervention study on acne comparing isotretinoin with control. Data are presented as percentage change from baseline, where negative values indicate a reduction over the trial. The sums of the isotretinoin and control sample sizes do not equal those reported in Table 1, as not all participants who were randomized completed the trials. Positive values for difference between groups indicates a greater reduction by isotretinoin than control

	Isotretinoin (ISO)		Control (CON)					
Study	n	Acne change from baseline, % (95% CI)	n	Acne change from baseline, % (95% CI)	Difference between groups, % (95% CI)	P-value	Acne change measure	Notes
Goldstein 1982[22]	28	−42·7 (−24·4 to −61·0)	28	−7·8 (2·1 to −17·7)	34·9 (14·1–55·7)	0·0026	Total lesion count	–
Gollnick 2001[23]	35	−90·3 (−80·5 to −100)	50	−77·3 (−65·7 to −88·9)	13·0 (−2·2–28·2)	0·11	Total lesion count	Calculated based on sum of median values for comedo, papule and nodule count
Jones 1984, 1989[24,25]	26	−76·6 (−64·2 to −88·9)	27	−57·9 (−43·5 to −72·3)	18·7 (−6·0–43·4)	0·15	Total lesion count	–
Lester 1985[26]	15	−73·4 (−51·1 to −95·8)	15	−52·7 (−27·5 to −78·0)	20·7 (−13·0–54·4)	0·24	Total lesion count	Calculated from sum of cyst, comedo and pustule count
Lin 1999[27]	8	−62·5 (−29·0 to −96·1)	10	−7·5 (8·8 to −23·8)	55·0 (17·7–92·3)	0·013	Acne severity scale	–
Oprica 2007[28]	24	−93·3 (−83·3 to −100)	25	−65·8 (−47·3 to −84·4)	27·5 (6·4–48·6)	0·018	Total lesion count	Calculated by converting log to linear scale, and then taking sum of superficial inflammatory acne, deep inflammatory acne and noninflammatory acne counts
Peck 1982[29]	16	−32·0 (−9·1 to −54·9)	17	58·0 (34·5 to 81·5)	90·0 (75·7–100)	< 0·001	Total lesion count	Placebo control
Prendiville 1988[30]	20	−87·3 (−72·7 to −100)	20	−47·5 (−25·6 to −69·4)	39·8 (13·5–66·1)	0·0073	Total lesion count	Calculated from sum of back and face lesion counts
Rademaker 2014[31]	29	−69·8 (−53·1 to −86·5)	29	−11·3 (0·2 to −22·9)	58·5 (38·2–78·8)	< 0·001	Total lesion count	Placebo control
Tan 2014[32]	133	−92·9 (−88·5 to −97·3)	133	−78·2 (−71·2 to −85·2)	14·7 (6·4–23·0)	< 0·001	Total lesion count	–
Wahab 2008[33]	30	Not possible to report	30	Not possible to report	Not possible to report	–	Global acne grading score	ISO: 24 with complete clearing, 5 with 75% clearing, 1 with 50–75% clearing. CON: 6 with complete clearing, 9 with 75% clearing, 3 with 50–75% clearing, 6 with < 50% clearing, 6 with no clearing

CI, confidence interval.

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Table 3 Summary of adverse event frequencies across studies. Data grouped by primary system affected due to inconsistent reporting across studies

System affected	Adverse event frequency	
	Isotretinoin	Control
Abnormal blood work	15	5
Elevated cholesterol, elevated liver enzymes, elevated triglycerides, reduced haemoglobin		
Dermatological	487	227
Acne flare, alopecia, cheilitis, crusting of lesions, decreased sweating, dermatitis, desquamation, dry hair, eczema, erythema, flushing, herpes, infection, morbilliform eruption, photosensitivity, pruritus, skin fragility, skin reaction, Stevens–Johnson syndrome, xerosis		
Ear, nose and throat	87	35
Dry mouth, dry nose, epistaxis, rhinitis, sore mouth		
Gastrointestinal	15	34
Abdominal pain, altered appetite, diarrhoea, indigestion, nausea, vomiting		
Ophthalmological	54	17
Blurred vision, conjunctival infection, conjunctivitis, dry eyes, irritation, pain, photophobia, pterygium		
Psychiatric	32	19
Depressed mood, fatigue, hallucination, insomnia, lethargy		
Other	61	51
Headache, increased thirst, musculoskeletal pain, tender fingertips, unknown		
Total	751	388

randomized to isotretinoin withdrew from trials due to elevated liver enzymes, while one participant (0·3%) randomized to a dapsone control group withdrew for this reason.

Dermatological events

The frequency of dermatological adverse events was over twice as high in the isotretinoin group than in controls, but both groups experienced a substantial number of adverse events. Dermatological adverse events represented 487 of 751 (64·8%) of all reported adverse events for those randomized to isotretinoin, with cheilitis and xerosis being very common. With regard to severe dermatological complications, one patient (one of 372, 0·3%) randomized to isotretinoin withdrew from the study and required hospitalization due to Stevens–Johnson syndrome. Other causes of study withdrawal with isotretinoin use included cheilitis (two of 372, 0·5%), xerosis (two of 372, 0·5%), acne flare (one of 372, 0·3%) and photophobia (one of 372, 0·3%). Control group withdrawals due to dermatological adverse events included skin reaction (one of 388, 0·3%) and morbilliform eruption (one of 388, 0·3%; the same patient who withdrew due to elevated liver enzymes from dapsone).

Ear, nose and throat

Ear, nose and throat adverse events were also twice as frequent in the isotretinoin groups than in controls. These adverse events represented 87 of 751 (11·6%) of all adverse events for those treated with isotretinoin and can largely be attributed to dryness of the oral and nasal mucous membranes. No participants withdrew from trials due to ear, nose or throat problems.

Gastrointestinal

Among all adverse event subtypes, gastrointestinal complications were the only ones more frequently reported in control groups. These reports were primarily from one study,[32] where nausea and vomiting were more frequent with antibiotic therapy (doxycycline and topical adapalene/benzoyl peroxide gel) than with isotretinoin. There were five of 388 (1·3%) control patients who withdrew from studies due to nausea, vomiting, diarrhoea or abdominal pain, but these were all in antibiotic groups. Importantly, no documented cases of inflammatory bowel disease were reported in either study group. One patient treated with isotretinoin (one of 372, 0·3%) withdrew due to decreased appetite. Gastrointestinal adverse events represented 15 of 751 (2·0%) adverse events for those treated with isotretinoin.

Ophthalmological

Ocular adverse events were again twice as frequent in the isotretinoin groups than in the control groups. These adverse events represented 54 of 751 (7·2%) of all adverse events for those treated with isotretinoin, where eye dryness, irritation and conjunctival infection were most commonly reported. No patients withdrew from studies due to ophthalmological problems.

Psychiatric and psychosomatic events

Psychiatric or psychosomatic adverse events were about 50% more frequent with isotretinoin use than in controls. Most commonly, this was documented as fatigue or lethargy. Overall, psychiatric or psychosomatic adverse events represented 32 of 751 (4·3%) adverse events for those treated with isotretinoin. Across all studies, there were two of 372 (0·5%) patients in the isotretinoin group who withdrew from the study due to psychiatric symptoms such as depressed mood, insomnia and hallucinations,[31,32] although none of these cases was reviewed by a psychiatrist for diagnosis. One additional case of depressed mood requiring study withdrawal was noted

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during the crossover phase of a placebo control group,[31] but as it is unclear whether this was due to isotretinoin therapy or acne worsening from treatment delay, it should be considered with caution. No other cases of study withdrawal due to psychiatric or psychosomatic symptoms were reported in the control groups.

Others

Other adverse events were balanced between the isotretinoin and control groups. Commonly, these included arthralgia and headache. There was one patient out of 372 (0·3%) randomized to isotretinoin who withdrew due to headaches; however, no cases of idiopathic intracranial hypertension were noted. Other adverse events represented 61 of 751 (8·1%) adverse events for those treated with isotretinoin.

Interpretation

This review has found evidence that oral isotretinoin can reduce acne lesion counts to a greater extent than placebo, oral antibiotics, etretinate or vitamin B complex. While the overall quality of evidence was low in this review, two of three studies evaluated as having an overall low risk of bias demonstrated clinical relevance and statistical significance when comparing isotretinoin with control. In addition, while adverse events were frequent with isotretinoin, severe adverse events requiring treatment cessation and study withdrawal occurred in only 12 of 372 (3·2%) cases.

This review supports the possibility that adverse event rates may be related to dose, consistently with previous literature.[35] Most commonly, adverse events were dermatological in nature. Isotretinoin has raised much discussion on its potential association with pancreatitis,[36] inflammatory bowel disease[18] and idiopathic intracranial hypertension;[37] however, these adverse events were not observed in this review of randomized controlled trials, possibly due to low sample sizes in the included trials. Evidence is still conflicting on the association between isotretinoin and psychiatric adverse events,[38] and the results from this review cannot rule out this possibility. We identified that psychiatric or psychosomatic symptoms were more frequent with isotretinoin therapy. Two participants withdrew due to depressed mood in the isotretinoin group during the randomized, blinded phase of the trials, and one participant withdrew due to depressed mood during an open-label crossover phase to isotretinoin. These cases were documented in studies that had well-matched groups at baseline, and so it is unlikely that these are due to confounding by acne severity. It should also be emphasized that these psychiatric events were documented in a short study period (< 20 weeks), and the long-term consequences are not known.

Only studies reporting randomized trials were considered. This was done on the basis that (i) nonrandomized trials cannot be used to evaluate causal effects of isotretinoin exposure,[39] and (ii) observational studies are prone to confounding by indication.[38] The concept of confounding by indication refers to the concern where, in nonrandomized trials, choice of treatment may be given in accordance with acne severity, and as a consequence it becomes impossible to discern whether outcomes and adverse events were due to the treatment provided or the severity of acne. Randomization is the only effective approach to ensure that all possible mechanisms that could connect acne severity with clinical outcomes are controlled. Consistently with the Canadian Clinical Practice Guidelines on Acne,[10] the American Academy of Dermatology clinical guidelines for management of acne[11] and the European Dermatology Forum guidelines for acne,[12] our review of evidence from randomized controlled trials supports the recommendation that oral isotretinoin should be considered for management of severe acne.

Limitations and future research

This review is limited primarily by a paucity of long-term follow-up data and significant heterogeneity in dosing, methodology, reporting and study samples across trials, making meta-analysis of data and assessment of acne recurrence unfeasible. Furthermore, there may be additional adverse events from isotretinoin documented in studies that were missed in this review. Study sample sizes were generally quite low – we identified only 760 participants involved in randomized controlled studies comparing isotretinoin with other therapies. Of these, fewer than 100 were in placebo-controlled studies and it is therefore possible that the sample sizes considered in this review were insufficient to observe all adverse events that may be associated with isotretinoin. Importantly, across all trials reported in this review, only 156 female participants were involved, largely due to study exclusion requirements to prevent teratogenicity in pregnancy,[40] and thus treatment efficacy and adverse event data from this review may not be as applicable to the female population. This is of concern regarding psychiatric adverse events; given that depression is more prevalent among women in the general population,[41] the overall incidence of this outcome is likely higher in clinical practice than in this review, given the low number of women studied.

Another important limitation is that the studies reviewed do not necessarily reproduce the manner in which isotretinoin is used in regular clinical practice. For instance, we were unable to assess the effects of different isotretinoin dosing strategies, sex, age or comorbidity on treatment efficacy or adverse events. Future research will need to focus on these factors to identify whether patient subgroups exist that are better suited or contraindicated for isotretinoin therapy, both from efficacy and adverse event perspectives.

Conclusions

This review provides support that oral isotretinoin can reduce acne lesion counts to a greater extent than control interventions. However, adverse events are more frequently seen with oral isotretinoin than with control, and may be more likely to

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occur at higher daily doses. Most isotretinoin adverse events are minor dryness-related skin symptoms, whereas adverse events from isotretinoin causing participant withdrawal from trials were limited to 3·2% of patients randomized to isotretinoin. These included Stevens–Johnson syndrome, severe cheilitis, severe xerosis, severe acne flare, photosensitivity, elevated liver enzymes, decreased appetite, headache and depressed mood. Consequently, the Canadian, American and European clinical guidelines on acne are well supported by this review.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1 Supplementary methods and search strategy.

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